

A counterexample to Erdős Problem 1075

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Abstract

Erdős asked whether, for every fixed $r \geq 3$, there is a constant $c_r > r^{-r}$ such that, for every $\varepsilon > 0$, every sufficiently large r -uniform hypergraph on N vertices with at least $(1 + \varepsilon)(N/r)^r$ edges contains a subgraph on $m \rightarrow \infty$ vertices with at least $c_r m^r$ edges. We give a counterexample for $r = 16$. We construct explicit finite hypergraphs G_n whose unnormalized Lagrangians satisfy

$$16^{-16} \left(1 + \frac{1}{4(14n)^{14}} \right) \leq \lambda(G_n) \leq 16^{-16} + \frac{1}{15!n}.$$

Blow-ups of these hypergraphs disprove the asserted density increase. The construction uses two families of bipartite links arranged around a cycle. A quantitative inequality for the corresponding open paths gives the upper bound.

1 Introduction

All hypergraphs in this paper are finite and simple. For a hypergraph H , write $v(H) = |V(H)|$, $e(H) = |E(H)|$, and let $H[S]$ denote the subgraph induced by $S \subseteq V(H)$.

Erdős proved that, for every fixed $r \geq 2$ and $\varepsilon > 0$, an r -uniform hypergraph on N vertices with at least εN^r edges contains a complete r -partite r -uniform hypergraph with equal part sizes tending to infinity with N [1]. In particular, it contains a subgraph on $m \rightarrow \infty$ vertices with $(m/r)^r$ edges. Erdős subsequently conjectured that, for each fixed $r \geq 3$, there is a constant $c_r > r^{-r}$ such that, for every $\varepsilon > 0$ and all sufficiently large N , the assumption

$$e(H) \geq (1 + \varepsilon)(N/r)^r$$

forces a subgraph on $m = m(N) \rightarrow \infty$ vertices with at least $c_r m^r$ edges [2, p. 80]. This is Problem 1075 in Bloom's collection [3].

Theorem 1. *For every $\gamma > 16^{-16}$, there is an $\varepsilon > 0$ and there are arbitrarily large 16-uniform hypergraphs H such that*

$$e(H) \geq (1 + \varepsilon)(v(H)/16)^{16}, \quad e(H[S]) < \gamma |S|^{16} \quad \text{for every nonempty } S \subseteq V(H).$$

The theorem disproves the conjecture at the fixed uniformity $r = 16$. It follows from a sequence of explicit finite hypergraphs with Lagrangians approaching 16^{-16} from above. We use the convention

$$P_H(x) = \sum_{e \in E(H)} \prod_{v \in e} x_v, \quad \lambda(H) = \max_{\substack{x_v \geq 0 \\ \sum_v x_v = 1}} P_H(x), \quad (1.1)$$

without the factor $r!$.

Proposition 2. *For every integer $n \geq 2$, the 16-uniform hypergraph G_n defined in Section 2 has $2n + 29$ vertices and satisfies*

$$16^{-16} \left(1 + \frac{1}{4(14n)^{14}} \right) \leq \lambda(G_n) \leq 16^{-16} + \frac{1}{15!n}. \quad (1.2)$$

The construction couples two three-uniform hypergraphs by a cyclic shift of their bipartite links. A prescribed weighting makes both cubic forms attain their separate upper bounds, and one additional edge gives a strict increase. When the additional edge has positive weight, cutting the cycle prevents simultaneous equality in the two cubic bounds. We quantify the resulting loss and use it to bound every open path. Deleting a pair of vertices of least total weight then gives the upper bound in (1.2).

2 The cyclic construction

Fix $n \geq 2$. Take pairwise distinct vertices

$$A_i, B_i \quad (i \in \mathbb{Z}/n\mathbb{Z}), \quad C, U, V, \quad W_1, \dots, W_{12}, \quad D_1, \dots, D_{14}.$$

Put $\mathcal{W} = \{W_1, \dots, W_{12}\}$ and $\mathcal{D} = \{D_1, \dots, D_{14}\}$. On the vertex set $\{A_i, B_i : i \in \mathbb{Z}/n\mathbb{Z}\} \cup \{C\} \cup \mathcal{D}$, define two three-uniform hypergraphs by

$$\begin{aligned} E(H_1) &= \bigcup_i \{ \{A_i, X, Y\} : X \in \{B_i, C\}, Y \in \{B_j : j \neq i\} \cup \mathcal{D} \}, \\ E(H_2) &= \bigcup_i \{ \{B_i, X, Y\} : X \in \{A_{i+1}, C\}, Y \in \{A_j : j \neq i+1\} \cup \mathcal{D} \}. \end{aligned}$$

All subscripts are read modulo n . Define

$$\begin{aligned} E(G_n) &= \{ \mathcal{W} \cup \{U\} \cup e : e \in E(H_1) \} \\ &\quad \cup \{ \mathcal{W} \cup \{V\} \cup e : e \in E(H_2) \} \cup \{ \{U, V\} \cup \mathcal{D} \}. \end{aligned} \quad (2.1)$$

Each edge has sixteen vertices. The three displayed families are disjoint, and the apex A_i or B_i determines the index within each link family, so the construction has no repeated edges.

Use lower-case letters for the weights of the corresponding vertices, and write

$$A = \sum_i a_i, \quad B = \sum_i b_i, \quad D = \sum_{j=1}^{14} d_j, \quad \omega = \prod_{j=1}^{12} w_j.$$

The two cubic forms and the polynomial of G_n are

$$P_1 = \sum_i a_i(b_i + c)(B - b_i + D), \quad (2.2)$$

$$P_2 = \sum_i b_i(a_{i+1} + c)(A - a_{i+1} + D), \quad (2.3)$$

$$P_{G_n} = \omega(uP_1 + vP_2) + uv \prod_{j=1}^{14} d_j. \quad (2.4)$$

Lemma 3. *The lower bound in (1.2) holds.*

Proof. Assign the weights

$$\begin{aligned} w_j &= \frac{1}{16}, & u &= v = \frac{1}{32}, & a_i &= b_i = \frac{1}{16n}, \\ d_j &= \frac{1}{224n}, & c &= \frac{n-1}{16n}. \end{aligned} \tag{2.5}$$

Their sum is one. Moreover,

$$A = B = c + D = \frac{1}{16}, \quad b_i + c = B - b_i + D = a_{i+1} + c = A - a_{i+1} + D = \frac{1}{16}.$$

Thus $P_1 = P_2 = 16^{-3}$. The first term in (2.4) is 16^{-16} , and the second is

$$\frac{1}{1024(224n)^{14}} = \frac{16^{-16}}{4(14n)^{14}}. \quad \square$$

3 A bound for open paths

We first prove an inequality for the cubic forms obtained by opening the cycle. The parameter D in the following lemma is a single nonnegative number; its division among fourteen vertices will be used only in Lemma 5.

Lemma 4. *Let $L \geq 1$, and let $a_1, \dots, a_L, b_1, \dots, b_L, c, D$ be nonnegative numbers. Put*

$$A = \sum_{i=1}^L a_i, \quad B = \sum_{i=1}^L b_i, \quad a_{L+1} = 0,$$

and suppose that $A + B + c + D = 1$. Define P_1, P_2 by (2.2) and (2.3), with sums over $1 \leq i \leq L$. Then, for every $s \in [0, 1]$,

$$sP_1 + (1-s)P_2 \leq \frac{1}{27} - \frac{s(1-s)D^3}{1200}. \tag{3.1}$$

Proof. Set

$$\delta_A = A - \frac{1}{3}, \quad \delta_B = B - \frac{1}{3},$$

and

$$\begin{aligned} \beta &= \frac{B + D - c}{2} = D + \frac{\delta_A}{2} + \delta_B, \\ \alpha &= \frac{A + D - c}{2} = D + \delta_A + \frac{\delta_B}{2}. \end{aligned}$$

Completing the square gives

$$P_1 = \frac{A(1-A)^2}{4} - \sum_i a_i (b_i - \beta)^2, \tag{3.2}$$

$$P_2 = \frac{B(1-B)^2}{4} - \sum_i b_i (a_{i+1} - \alpha)^2. \tag{3.3}$$

For $0 \leq x \leq 1$,

$$\frac{1}{27} - \frac{x(1-x)^2}{4} = \frac{(x-1/3)^2(4/3-x)}{4} \geq \frac{(x-1/3)^2}{12}.$$

Consequently, if

$$E = s \sum_i a_i (b_i - \beta)^2 + (1-s) \sum_i b_i (a_{i+1} - \alpha)^2,$$

then

$$\frac{1}{27} - sP_1 - (1-s)P_2 \geq \frac{s\delta_A^2 + (1-s)\delta_B^2}{12} + E. \quad (3.4)$$

This proves the assertion when $D = 0$ or $s \in \{0, 1\}$. Assume henceforth that $D > 0$ and $0 < s < 1$.

If $|\delta_A| \geq D/10$, the right side of (3.4) is at least

$$\frac{sD^2}{1200} \geq \frac{s(1-s)D^3}{1200},$$

since $D \leq 1$. The case $|\delta_B| \geq D/10$ is analogous. It remains to consider $|\delta_A|, |\delta_B| < D/10$. In this case

$$D \leq c + D = \frac{1}{3} - \delta_A - \delta_B \leq \frac{1}{3} + \frac{D}{5},$$

so $D \leq 5/12$. It follows that

$$\frac{17D}{20} \leq \alpha, \beta \leq \frac{23D}{20} \leq \frac{23}{48}, \quad A, B \geq \frac{1}{3} - \frac{D}{10} \geq \frac{7}{24}. \quad (3.5)$$

In particular, $\alpha, \beta > 0$, $A \geq \alpha/3$, and $B \geq \beta/3$.

Consider the finite sequence

$$\frac{a_1}{\alpha}, \frac{b_1}{\beta}, \dots, \frac{a_L}{\alpha}, \frac{b_L}{\beta}, 0. \quad (3.6)$$

Each successive pair z, z' contributes to E a term

$$s\alpha\beta^2 z(z' - 1)^2 \quad \text{or} \quad (1-s)\beta\alpha^2 z(z' - 1)^2,$$

according as the first entry comes from an a_i or a b_i . If some entry of (3.6) is at least $1/3$, then, because the last entry is zero, some successive pair satisfies $z \geq 1/3$ and $z' < 1/3$. Hence

$$E \geq \frac{4}{27} \min\{s\alpha\beta^2, (1-s)\beta\alpha^2\}. \quad (3.7)$$

If all entries are less than $1/3$, then instead

$$\begin{aligned} E &\geq \frac{4}{9} (s\beta^2 A + (1-s)\alpha^2 B) \\ &\geq \frac{4}{27} (s\alpha\beta^2 + (1-s)\beta\alpha^2), \end{aligned}$$

which also implies (3.7). Using (3.5), we obtain in either case

$$E \geq \frac{4}{27} \left(\frac{17}{20}\right)^3 \min\{s, 1-s\} D^3 \geq \frac{s(1-s)D^3}{1200}.$$

The last inequality follows from $\frac{4}{27} \left(\frac{17}{20}\right)^3 = 4913/54000 > 1/1200$. Together with (3.4), this proves the lemma. $\square$

For $L \geq 1$, define the 16-uniform hypergraph T_L as in (2.1), with each index ranging over $1, \dots, L$ and with the last link in H_2 replaced by the triples

$$\{B_L, C, Y\}, \quad Y \in \{A_1, \dots, A_L\} \cup \mathcal{D}.$$

The edge $\{U, V\} \cup \mathcal{D}$ is retained. Its polynomial is (2.4) with $a_{L+1} = 0$.

Lemma 5. *For every $L \geq 1$, we have $\lambda(T_L) \leq 16^{-16}$.*

Proof. Take nonnegative vertex weights of total mass one, and put

$$h = \sum_{j=1}^{12} w_j, \quad q = u + v, \quad t = A + B + c + D.$$

Thus $h + q + t = 1$. If $q = 0$ or $t = 0$, then $P_{T_L} = 0$. Otherwise put

$$s = u/q, \quad d = D/t, \quad z = s(1-s)d^3 \in [0, 1/4].$$

Applying Lemma 4 to the bottom weights divided by t , and then using homogeneity, gives

$$uP_1 + vP_2 \leq qt^3 \left(\frac{1}{27} - \frac{z}{1200} \right).$$

The arithmetic–geometric mean inequality now yields

$$P_{T_L} \leq \left(\frac{h}{12} \right)^{12} qt^3 \left(\frac{1}{27} - \frac{z}{1200} \right) + q^2 s(1-s) \left(\frac{td}{14} \right)^{14}. \quad (3.8)$$

Also,

$$\frac{qt^3}{27} \leq \left(\frac{q+t}{4} \right)^4, \quad \frac{q^2 t^{14}}{14^{14}} \leq 4 \left(\frac{q+t}{16} \right)^{16}. \quad (3.9)$$

The second inequality applies the same inequality to two copies of $q/2$ and fourteen copies of $t/14$.

Define

$$F(h) = \left(\frac{4h}{3} \right)^{12} [4(1-h)]^4.$$

We have $F(h) \leq 1$, with equality at $h = 3/4$. Since $1 - 27z/1200 > 0$, we may substitute (3.9) into (3.8). Using $d^{14} \leq d^3$, we get

$$\begin{aligned} \frac{P_{T_L}}{16^{-16}} &\leq F(h) \left(1 - \frac{9}{400}z \right) + 4s(1-s)d^{14}(1-h)^{16} \\ &\leq F(h) + z \left[4(1-h)^{16} - \frac{9}{400}F(h) \right]. \end{aligned} \quad (3.10)$$

Suppose first that $1/2 \leq h < 1$. Then

$$\frac{(1-h)^{16}}{F(h)} = \frac{(3/4)^{12}}{256} \left(\frac{1-h}{h} \right)^{12} \leq \frac{1}{256}.$$

Since $1/64 < 9/400$, the expression in brackets in (3.10) is nonpositive. Thus the right side is at most $F(h) \leq 1$. If $h = 1$, the polynomial is zero.

Suppose next that $0 \leq h \leq 1/2$. Discarding the negative term in (3.10) and using $z \leq 1/4$, we obtain

$$\frac{P_{T_L}}{16^{-16}} \leq F(h) + (1-h)^{16}. \quad (3.11)$$

For $0 < h \leq 1/2$, the function $F(h)/h$ is increasing, since its logarithmic derivative is

$$\frac{11}{h} - \frac{4}{1-h} = \frac{11-15h}{h(1-h)} > 0.$$

Moreover,

$$\frac{F(1/2)}{1/2} = 32 \left(\frac{2}{3}\right)^{12} = \frac{131072}{531441} < 1.$$

It follows that $F(h) \leq h$ throughout $[0, 1/2]$. The right side of (3.11) is therefore at most $h + (1-h)^{16} \leq 1$, as required. $\square$

4 The Lagrangian bound and the blow-ups

Proof of Proposition 2. The lower bound was proved in Lemma 3. For the upper bound, take any nonnegative weights on G_n of total mass one. Choose an index k with

$$a_k + b_k \leq \frac{A+B}{n} \leq \frac{1}{n}.$$

Deleting A_k, B_k leaves an induced copy of T_{n-1} : order the remaining indices as $k+1, \dots, k+n-1$ modulo n . The last B -vertex loses its successor A_k , giving precisely the terminal link in the definition of T_{n-1} . By Lemma 5 and homogeneity, the total weight of the edges that remain is at most

$$16^{-16}(1 - a_k - b_k)^{16} \leq 16^{-16}.$$

For any simple 16-uniform hypergraph and weights of total mass one,

$$\frac{\partial P_H}{\partial x_v} \leq \sum_{\substack{S \subseteq V(H) \setminus \{v\} \\ |S|=15}} \prod_{w \in S} x_w \leq \frac{(\sum_{w \neq v} x_w)^{15}}{15!} \leq \frac{1}{15!}. \quad (4.1)$$

Indeed, each product of fifteen distinct weights occurs $15!$ times in the power expansion, and all other terms are nonnegative. The deleted edges consequently have total weight at most

$$a_k \frac{\partial P_{G_n}}{\partial a_k} + b_k \frac{\partial P_{G_n}}{\partial b_k} \leq \frac{a_k + b_k}{15!} \leq \frac{1}{15!n}.$$

Edges containing both deleted vertices are counted twice, which does not affect the upper bound. This proves (1.2). $\square$

Proof of Theorem 1. Fix $\gamma > 16^{-16}$, and choose an integer $n \geq 2$ such that

$$16^{-16} + \frac{1}{15!n} < \gamma.$$

Keep n fixed, and set

$$\varepsilon = \frac{1}{4(14n)^{14}} > 0.$$

For each positive integer M , form a blow-up $B_{n,M}$ of G_n : replace each vertex by an independent class, and each edge by all transversal edges across its sixteen classes. Use the following class sizes, with one class for each listed vertex:

vertex	W_j	U, V	A_i, B_i	C	D_j
class size	$14nM$	$7nM$	$14M$	$14(n-1)M$	M

There are $N = 224nM$ vertices. These class sizes are proportional to the weights in (2.5), so Lemma 3 gives the exact edge count

$$e(B_{n,M}) = [(14n)^{16} + 49n^2]M^{16} = (1 + \varepsilon)(N/16)^{16}. \quad (4.2)$$

Let S be any nonempty set of m vertices of $B_{n,M}$, and let k_v be the number of its vertices in the class corresponding to $v \in V(G_n)$. Then

$$e(B_{n,M}[S]) = \sum_{e \in E(G_n)} \prod_{v \in e} k_v = m^{16} P_{G_n}((k_v/m)_v) \leq m^{16} \lambda(G_n) < \gamma m^{16}.$$

Together with (4.2) and $M \rightarrow \infty$, this proves the theorem. A non-induced subgraph has no more edges than the induced subgraph on the same vertex set. $\square$

AI tool disclosure

OpenAI Codex (GPT-6) was used substantially in developing this work. It proposed the cyclic construction, derived the quantitative path inequality and the Lagrangian estimates, and drafted the manuscript. It also assisted with literature searches and exact algebraic checks. The author specified the research problem and directed the preparation of the paper, and is responsible for its mathematical content.

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